National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Data Analysis and Visualization	Course Code:	DL3001
Program:	BS Data Sciences	Semester:	Fall 2023
Section	BS DS 5A	Total Marks:	15
Date:	29th November 2023	Weight	
Exam Type:	Quiz 4	Page(s):	2

Q1: Answer the question keeping in view the provided scenario. **[5+5 Marks]**

Scenario:

You are a data analyst for a retail company, and the management has provided you with the sales data for the past year. The data includes information on monthly sales figures, product categories, and customer demographics. Your task is to create visualizations to communicate key insights to different stakeholders within the company.

(a) Considering the product categories in the dataset, what type of visualization would effectively communicate the sales distribution across different product categories?

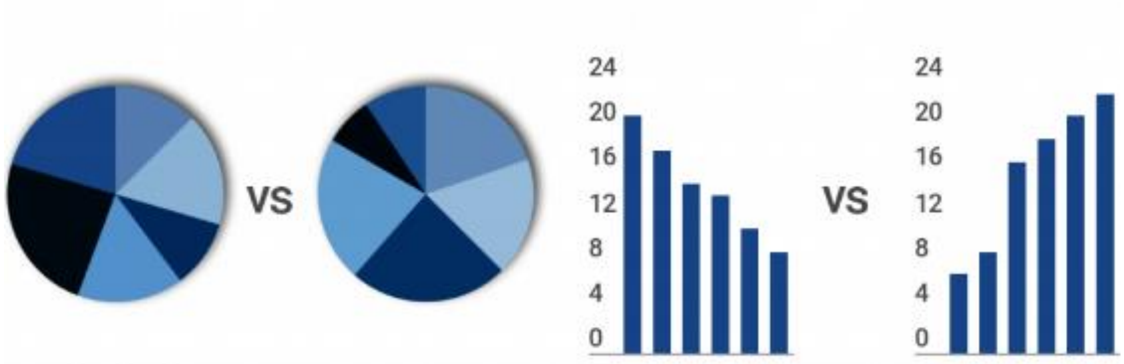
A stacked bar chart or a grouped bar chart can effectively communicate the sales distribution across product categories, allowing easy comparison of sales performance for each category.

(b) What type of graph or chart would you use to present an overview of the total sales for each month over the past year? Why?

I would use a line chart to display the trend in total sales over the past year. This type of chart is effective for showing changes over time, and it provides a clear visual representation of monthly sales fluctuations.

Q2: Consider the following visualizations and determine which one is correct for comparing two datasets side by side. Explain your answer.

[5 Marks]



To compare two datasets side by side we use bar and column charts instead of pie charts. It's much easier for people to compare the length of bars and columns than it is for them to compare the angles found in a pie chart. When comparing two datasets, a grouped bar or column chart is used. This means you don't have two separate charts to compare. Instead, you have one organized chart that saves space and your viewer's time. Therefore, the right visualization is correct.